Each attribute and its corresponding type are defined by the attr function. The type can either be a struct or one of a few predefined types such as :string, :integer, :any, or others.

In addition to the type of attribute, we can define whether it's required as well as a default value. Our form always requires an Ecto.Changeset struct, but the contact attribute is allowed to be nil. There are a few other attribute options[24] that you can use. These options serve as compile-time warnings and provide documentation.

If you start your application, you'll see that a warning now appears for the message_html/new.html.heex file:

```
warning: attribute "contact" in component
  PhoneAppWeb.MessageHTML.message_form/1 must be a
  {:struct, PhoneApp.Conversations.Schema.Contact}, got: nil

lib/phone_app_web/controllers/message_html/new.html.heex:6
```

To fix the warning, you can remove the contact: nil attribute from the noted line (new.html.heex:6).

We get this warning because we're passing nil in for a component that must be a Contact struct. This was accepted by the compiler before we defined the attribute as having a specific type. Once we define its type as a Contact struct, the compiler provides us with this helpful warning message.

The message_form component is about as basic as it gets. It takes a few known inputs and renders HTML based on them. But components can actually do more than that. Let's look at some of the more advanced features.

Advanced Component Features

Our app is pretty basic, so we don't have any complex components in it. But there are a few features that are useful as you're building out your component library: ~H sigil and slots.

The ~H sigil lets you embed the HTML for a component directly in an Elixir function. This is largely a matter of personal preference, but the embedded template style can be a bit easier to work with. To use it, you simply return an ~H wrapped string from your component function.

Here's a basic button component:

24. https://hexdocs.pm/phoenix_live_view/Phoenix.Component.html#attr/3